Gauri Agrawal

Software Engineer | Machine Learning

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Open to relocation nationwide

Education

Texas A&M University College Station, TX

B.S. in Computer Science; Minors in Mathematics, Game Design and Development *Aug. 2023 – May 2027*

– GPA: **3.797/4.0** | Lechner Scholar | Engineering Honors

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, C#, HTML/CSS, Haskell, x86 Assembly, R

Frameworks/Libraries: React, Vite, Tailwind CSS, FastAPI, Flask, Express/Node.js, SQLAlchemy

Machine Learning/AI: PyTorch, Graph Neural Networks (GNN), Conditional VAEs, Retrieval-Augmented Generation (RAG), OpenAI APIs

Systems/Infrastructure: Docker, AWS, Linux/WSL, PostgreSQL, Alembic, MPI, Slurm, RISC-V, Computer Architecture

Developer Tools: Git, JWT, Clerk, REST APIs, Agile/Scrum

Experience

Machine Learning Researcher, Data Integration, Visualization, and Exploration Lab College Station, TX

Texas A&M University *May 2026 – Present*

– Developed a **GNN-based CFD surrogate** predicting fluid-flow variables during Mars atmospheric reentry from graph-structured simulation data and physics-informed Mach/Reynolds features.

– Reduced 746-step autoregressive rollout error by **99.9% (∼1,800×)** against a training-budget-matched baseline using curriculum pushforward training.

– Scaled training across angles of attack using **METIS graph partitioning, MPI, and Slurm**; developed a Conditional VAE for collision trajectory generation (see Projects).

Curriculum Developer & Teaching Assistant College Station, TX

Texas A&M University – Dept. of Computer Science *Aug. 2025 – Aug 2026*

– Redesigned the *CSCE 312* lab curriculum around **RISC-V**, creating an 8-lab progression from digital logic to a working CPU and authoring lab manuals, solutions, and rubrics.

– Supported **300+ students** in *CSCE 222* through grading, office hours, and review sessions on discrete mathematics and proofs.

Software & Computer Vision Engineer, TAMU Karura College Station, TX

Texas A&M University *Jan. 2026 – Present*

– Benchmarked 5 object detection models across camera configurations using a reproducible evaluation pipeline, selecting **RF-DETR-N** at 0.84 mAP@50, a **32% improvement** over the best YOLO baseline (0.63).

– Achieved **2–3× higher mAP@50 on small objects** than YOLO baselines and 0.95 mAP@50 with 97% recall for mallet detection, improving rover tool recognition.

– Evaluated camera resolution trade-offs, maintaining mAP@50 within 0.01 from 720p to 480p while reducing transmitted pixels by **55%** to support bandwidth-efficient rover perception.

– Deployed object detection models using **TensorRT** on NVIDIA Jetson hardware and developed GUI screens integrating camera feeds for rover monitoring and interaction.

Projects

PaperTrail | *React, TypeScript, FastAPI, PostgreSQL, Docker* Jul. – Aug. 2026

– Architected a full-stack research paper management app: React/TypeScript frontend and a layered FastAPI backend for ingestion, metadata, and AI-assisted retrieval.

– Implemented **Clerk authentication**, JWT verification, and per-user data isolation for secure multi-user access.

– Built a PDF ingestion pipeline (pdfplumber, sentence-aware chunking) feeding a retrieval-augmented generation system.

– Integrated CrossRef API for DOI metadata and PostgreSQL/Alembic migrations; wrote a **40+ test** automated suite and Docker Compose environment.

N + N_2 Collision Trajectory Generation | *PyTorch, Python, Conditional VAE* June 2026

– Developed a two-phase generative ML framework (residual MLP + Conditional VAE) to emulate quasi-classical trajectory simulations for reactive N + N_2 collisions.

– Trained on **5.47M+ trajectories across 91,810 initial conditions**; validated generalization across 10 held-out energy levels and implemented counterfactual trajectory generation.

Boba Bliss – Point-of-Sale System | *Flask, PostgreSQL, React, Express/Node.js* Oct. – Nov. 2025

– Led a **4-person team** as Project Lead/Scrum Master, building a full-stack POS system (Flask + PostgreSQL backend, React frontend) supporting menu management, order processing, and role-based permissions.

Honors & Awards

2nd Place, Texas A&M Student Research Week – Computational & Data Sciences, Undergraduate (2025), for AI-assisted stroke rehabilitation guidance research | ECSM 2025 Presenter